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SUPERSEDING
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MILITARY SPECIFICATION

ION EXCHANGE COMPOUNDS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic macroreticular or macroporous cation and anion exchange compounds.

1.2 Classification. Ion exchange compounds shall be of the following types, as specified (see 6.2):

Type I - Strongly acidic cation exchange compound in the hydrogen form.
Type II - Weakly basic anion exchange compound in the free base form containing only tertiary amine functionality.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 4610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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AMERICAN WATER WORKS ASSOCIATION (AWWA)

B250 - Cation Exchanger Test Procedures.

(Application for copies should be addressed to the American Water Works Association, 2 Park Avenue, New York, NY 10016.)

3. REQUIREMENTS

3.1 Description. The ion exchange compounds are synthetic macroreticular or macroporous cation and anion exchange compounds used in water purification units.

3.2 Material. The ion exchange compounds shall be unused, wet synthetic compounds in bead form. The compounds shall have a macroreticular or macroporous and strain-free structure and be resistant to breakdown on chemical oxidation and wet-dry cycling. The compounds shall conform to Table I, be thoroughly rinsed, and be ready for use in a regenerated condition.

3.3 Unit content. The unit content shall be in cubic feet of ion exchange compound, as specified (see 6.2).

TABLE I. Quantitative requirements.

	Cation compound		Anion compound	
	Minimum	Maximum	Minimum	Maximum
Effective size, mm.	0.40	0.55	0.40	0.55
Uniformity coefficient	-	2.0	-	2.0
Compound passing 50-mesh sieve, percent by volume	-	2.0	-	2.0
Compound retained on 16-mesh seive, percent by volume	-	5.0	-	5.0
Whole bead content, percent	90.0	-	90.0	-
Wet density, lbs/cu ft.	46.0	52.0	37.0	42.0
Solids content, as received, percent by weight	45.0	-	40.0	-
Moisture holding capacity, percent by weight	51.0	56.0	46.0	52.0
Conversion to hydrogen form (as received) percent	90.0	-	N/A	N/A

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TABLE I. Quantitative requirements. - Continued

	Cation compound		Anion compound	
	Minimum	Maximum	Minimum	Maximum
Total exchange capacity, *Meq/gram	4.30	N/A	N/A	N/A
*Meq/milliliter, wet compound	1.60	N/A	N/A	N/A
Weak base capacity, *Meq/gram	N/A	N/A	4.20	-
Strong base capacity, *Meq/gram	N/A	N/A	-	20 percent of total capacity
Oxidative stability, percent moisture holding capacity after oxidative attack	-	65.0	-	-
Resistance to breakdown on wet-dry cycling, percent resin passing 50-mesh sieve	-	3.0	-	3.0

* Milliequivalents

3.4 Water soluble impurities. Nonvolatile impurities in the water extract shall constitute less than 0.10 percent of the resin when tested as specified in 4.3.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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